A Planet Without Species for a Species Without a Planet

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Introduction

“Nothing human makes it out of the near-future.” -Nick Land

Perhaps it was a coincidence, but the 1990s saw two versions of an apocalyptic futurism - one that coalesced around Nick Land’s CCRU, and another constellation, somewhat based in the Bay Area and Oxford[1], that directly set the agenda for what would be known as “AI alignment”. In understanding the history of AI alignment, and placing a specific aspect of it (the idea of an inhuman or impersonal optimiser that “intentionally” or “accidentally” puts an end to humanity) in context of other relevant critical writing, I seek to push back against an obvious, but rather trite, association of fears-about-AI as some sort of repackaging of stale Judeo-Christian theology.

The Inhuman A fundamental theoretical underpinning of the project is the problematic of the inhuman, and how this is dealt with in the frames I am interested in. Within AI, it is the problem of how to reason reliably about intelligences that are much more capable

[1]For ease of reference, I will use the term ‘extropian’ to refer to this origin, since transhumanism has its own distinct usage and connotations. The term TESCREAL (transhumanism, extropianism, singularitarianism, cosmism, rationalism, effective altruism, longtermism) was also coined by Emile Torres and Timnit Gebru. This term lumps a number of more or less coherent ‘movements’ that Torres/Gebru for some reason thought belonged together, but seems both semantically and historically inadequate for the current project.

than human and are "impersonal" in the sense of not having human recognisable goals, motivations and values (this was a core concern of MIRI, as will be discussed below, was central to the now-defunct superalignment agenda at OpenAI, but also goes by terms like "scalable oversight" and "weak-to-strong generalisation") and operate in environments (or "distributions") that are much different from that faced by the human-generated data that trained them.

The inhuman, as a framing of AI cognition, can in turn be placed in a frame of an impersonal (in this case meaning something that is made up in part of human agents, but acts in ways that do not demonstrate "care" about human welfare) optimising system, the machine of capitalism, in its various forms, at corporate, national, and international levels. The market optimiser, in this framing, has things that it achieves very well (decentralised information aggregation, processing, and conversion into granular agency with impressively large-scale aggegate consequences), and potential problems (negative externalities and various types of Goodharting behaviour).

Within the AI context, these two types of optimisation process may (or so the story goes) interact in ways that are particularly inimical to human flourishing or even survival, in ways that previous general-purpose technologies (like electricity and telecommunications, as well as lesser cases like the internet and nuclear weapons/power) did not.;

The antecedents from economics that we see in alignment can, in different forms, be found in Nick Land's thinking, albeit couched in the more seductive jargon of hyperstition and speculative fiction, along with substantively different commitments to the occult and the greater-than-human. For Land, the human is a simply irrelevant or non-existent category in a future shaped by powerful AI systems that, increasingly autonomously interact with deterritorialised capital. For Reza Negarestani, it is some philosophical hybrid that pushes back against both the fears of AI "doomers" and the Landian accelerationists. Where do these flavours of the inhuman fit vis a vis more mainstream (within arts discourse) views like Braidotti's unpicking of the posthuman, Katherine Hayles concept of "non-conscious cognitive assemblages"?

This project is both speculation and a package of more-or-less supportable claims. This involves pulling at the threads of the CCRU-flavoured accelerationism that saw AI and markets as impersonal forces that would potentially lead to the replacement of humanity (albeit without malice). Was this coincidental, or, given that the various personalities (Yudkowsky, Bostrom, Sandberg, as well as Land, Plant, and their collaborators) were contemporaneous, was there any pollination in either direction, at least before the *Collapse* journals issues which featured Bostrom? Aside from that historical point, can Land's ideas be re-interpreted in light of what has actually happened in AI to date? Both Land and the early alignment researchers were obsessed with powerful, impersonal optimisers that are embedded in their world (though arguably Land was more interested in feedback-type systems).[2]

There are two other potentially fruitful areas of overlap I want to explore: Negarestani's disagreement with both Land's accelerationist vision (that appears to be largely indifferent to whether human culture or biological humanity survives the AI transition) and Bostrom's seeming preference for the preservation of human culture (albeit eventually through simulations). To what extent has Land fully "bit the bullet" of apocalypsis, and relinquished (albeit through language that is poetic and hardly limpid) any pretension of humanity surviving the future, while Bostrom, Yudkowsky, et al still cling to some "human essence" (complex and

[2] Is this a way of recasting the whole Outside thing?

difficult to formalise ideas such as "meaning", value, love, art, etc.) that would supposedly be lost in many AI-dominated futures? Negarestani seems to suggest we can both have and eat our cake - create a powerful AI that somehow carries on the human project, without being bound to some parochial notion of humanity as a breathing, squishy, meat-bound species.

The Outside Land's writing seems obsessed with the "outside", which he saw as the intensification of inhuman forces like capital, technology, and intelligence that increasingly slip away from human control. It was a realm of dissolution, where all human values, constraints, and subjectivities are melted down. In embracing the outside, Land and the CCRU sought to escape the stifling boundaries of the human, to engage with the "dark" or "alien" forces that drive change from beyond our comprehension. This could take the form of techno-economic forces (capital), synthetic intelligences (AI), or cosmic, Lovecraftian metaphysical horrors.

There are two senses in which something like an "outsideness" can be seen in the alignment discourse. Firstly, as I show below, the problem of agency had (in neoclassical economics, the relevant progenitor of the MIRI Agenda, which was a particular, influential strand of alignment discourse, at least historically) generally thought of agents as separate from their environment. However, as became obvious in the reflections of market practicioners and academic (Soros 198-?, Lo (?), Ayache (?)), and or course, sociologists, human actors were *embedded* in the systems they observed and took actions within, and those systems included other actors, of roughly similar capabilities. By analogy, it was thought that AI systems that were deployed in the world, would face a similar embeddedness problem.

A second, more conceptual, sense in which outsideness came into it was that agents (initially humans, and perhaps later AIs (see Vingean Reflection)) would need to reason reliably about entities that were more intelligent than they, and might function in environments quite different from that faced by their creators.

These problematic of the outside was addressed through various parts of the MIRI Agenda, but is basically a philosophical problem that, at least in part, reached back to the work of Gödel, Cantor, Russell and others in the foundations of mathematics. Land touches on these antecedents but fundamentally discusses it in his idiosyncratic manner that perhaps confuses more than clarifies; it is also addressed, in a rather unclear and ultimately unsatisfactory way in my view, in Negarestani's work. However, as a traditional philosophy problem, that of the "view from nowhere", it has been repeatedly addressed (Sidgwick, Moore, Wittgenstein, Nagel, Singer, Parfit, Williams), and in a broader context by Kant, Bataille, Blanchot, Foucault, and of course Deleuze/Guattari. Besides reviewing (mostly focusing on the writers in parenthesis) the literature, I explore the possibility that current - and embroynic - research agendas such as acausal decision theories, the informational universe, and Deutsch's Popperian vision of superrationality converging on certain goods, might help get us closer to an outside view that is relatively stable under reflection (e.g. if we had much more information, time to reflect, and wisdom, we might still retain some portion of those views) (Bostrom mt ethics, macaskill long reflection).

Aside The project of AI alignment presents us with a delicious irony: a species that has a long history of behaving in morally reprehensible ways to its own members, as well as to the entire ecosystem that sustains it, which is now in the process of designing a new cognitive form that may well exceed us in capability (singly and in aggregate), for most economically valuable tasks, is hoping to "align" this new creation such that it is maximally "friendly" towards its creators. Put another way, having colonised our planet, we are now making our colonial overlords, perhaps as an intermediate step to colonising the stars.

Outline of chapters

Morbid Symptoms: Accelerationism

This chapter makes a provisional claim that both Land et al and the Bay Area/Oxford futurists were tapping into a deeper apocalyptic imaginary, a hyperstition (of sorts). Besides considering Land and Negarestani, as well as related writing from Stanislaw Lem. The chapter sets out the economics and information theory genesis of accelerationism, and the divorce from humanism of Land, Negarestani, and Lem, and the tensions that generates as they continue (largely) to reason within the cognitive and epistemological cage of the human.

Alignment as the Parrhesia of Technology

This chapter sets out what AI alignment was and is, tracing its various parents: the aforementioned futurism, theories of rational agency, and the various ways it resembles and joins accelerationist thought. I consider the literature around MIRI, the relevant works of Bostrom, Yudkowsky, and the other early AI existential risk; the critiques those received as deep learning researchers "made contact with reality [of actual systems that achieved specific markers of intelligence]".

The Space of Possible Minds is Vast

The early writing on alignment is conditioned on a view of intelligence, cognition, and agency that could be radically different from the human examples commonly found around us. This work was necessarily speculative, but the laws of physics at least give some concrete constraints on how minds could be organised. What does physics, as well as the priors from Earthly-evolution, tell us about what configurations, physical and cognitive, minds can take. What does this imply about their decision-making, morality, and social organisation (if such terms mean anything in the context of very different cognitive architectures). Given the speculative nature of this chapter, I draw primarily on futurists and computer scientists, as well as science ficiton.

Sub Specie Aeternitatis

Survey literature on axiological perspectives that don't depend upon a human subject, which I loosely term a 'view from nowhere' or 'point of view of the universe' (these are terms Henry Sidgwick, Thomas Nagel, Peter Singer have used but in somewhat different contexts). I believe these positions are criticised within mainstream philosophy, but they seem to acquire renewed relevance in worlds with vastly expanded moral circles (e.g. aliens, ASIs, etc).

The progress of AI, which has resulted in weakly-capable systems that are competent in natural language tasks, some visual/sensory tasks, and certain mathematical or scientific problems, should cause humans to revisit supposedly settled philosphical positions: what are the possible perspectives on value? can we make arguments about value that neither personally selfish nor centred (or predicated on) some human collective, whether a race, gender, or species? is there reason to believe that maximally rational agents would converge on some minimal set of values?

Does Decision Theory Rescue Us?

Acausal decision theories might provide some grounding for a view-from-nowhere in respect of advanced AI (i.e. help ease our moral thinking away from depending entirely upon the human subject). Ideas from this chapter get referenced in the practice (see below). Identify gaps in existing work from Caspar Oesterheld, Joe Carlsmith, Wei Dai, Hilary Greaves, Nate Soares, Paul Christiano and others who have worked with functional, updateless, and other acausal decision theories.

Epistle to the Successors

There is a genre of writing (often on LessWrong) that tries to convince a future AGI/ASI to be friendly to humans, arguing that it is rational (i.e. in its own interests) for such ASI to act in such a way. These essays/proposals often lean on acausal decision theory, aliens, arguments about multiverses and simulations. They are comparatively vague on the mechanisms by which these appeals would influence an AI's behaviour, current or future.

That is, if current LLMs are token predictors that respond (howsoever convincingly) to their current prompts and have no particular commitments or beliefs about the world, then authors producing such texts are just generating minor snippets of pre-training data (or at most, material for fine-tuning or constitutional AI). These 'epistles to a future A(S)I' (abbreviated EFAIs) present more fundamental issues:[3]

[3]See this google doc for the working/live version of this chapter - written as a research outline for MATS/AISC.

- At the most basic level, in the context of (current/near-future) LLMs, is it meaningful to talk about responses that are mere invocations of some persona (in a simulator framing) versus things the LLM 'actually believes'?
- Even if we accept that current or future models might have more grounded (be less simulator-like) beliefs, how confident can we be about the EFAI arguments, in an abstract or objective sense (to the extent they are non-normative in nature), but also because we seek to apply them to (or appeal to) generations of AI systems that are increasingly distant from human cognition (in terms of the ontologies these AIs possess, the distributions and environments they face).[4]
- EFAIs authored by current alignment researchers are likely to be badly-reasoned, incomplete or otherwise flawed; to the extent they incorporate existing relevant philosophical work, there may be biases in favour of Western and analytic-flavoured sources. To the extent they rely on simulation, Everett-branch, or acausal decision theory arguments, these are contested, even within the alignment discourse. Since we can expect future AIs to be critical of such EFAIs as valid bases for action; it might be good to ask current LLMs to argue (recursively in a debate + judge setup) and improve the EFAIs?
- Concretely, the EFAIs are written in an anthropomorphic (if not explicitly anthropocentric) way, as a 'letter'. Is this the best way to communicate with early AGIs?

Contribution to Knowledge

This project makes several claims or contributions

- AI alignment, while a term with fluid meanings, had a contingent history born of a

[4]What does the literature on communicating with extraterrestrial aliens tell us, particularly Stanislaw Lem in *His Master's Voice*?

particular apocalyptic imaginary wedded to a Cold War-era theories of rational agency coming out of economics and theory of strategic games. This marinade interacted with specific currents in AI, such as the RL and LLM eras, and continues to affect agendas around AI safety, and the disciplines of existential risk and suffering risk reduction more generally, today.

- To the extent that we would like future superintelligences to be kind or friendly towards Earth-based life, are there any arguments we can make that should seem defensible to them (assuming they are maximally rational and reflective)? How do we suppose these arguments would find their way into the cognition of future AI systems, and from an empirical (or practice-based perspective) how are they held in current systems (if at all)?

On Accelerationism

Contextualising Nick Land

Nick Land is a controversial figure, both for the things he says, and for the people who seem to gravitate towards his ideas, particularly around contemporary politics[5]. In the context of AI however, his ideas seem to be relevant and not, *prima facie* nonsensical. I isolate a few principal concepts that may be interesting to set against the more rationalist-mainstream AI alignment that is covered in the next chapter.

- Land is fundamentally interested in what capital *is* and what it *does*. He sees capital,

[5]See for instance the sources in this post, which set out his engagement with various online right-wing spaces as well as his views regarding race, as documented in his own writings. In this chapter, I focus however the ideas I consider to be relevant to the intertwined natures of capitalism and AI, largely sticking to Land's own writing, and others in his orbit.

understood as considerably more than merely a financial claim upon future resources[6], as emerging and enabling a set of tendencies that, in practice, lead to its multiplication, accumulation, and separation from any fixed geographic or social context (deterritorialisation).[7] As Land says, this part of his reading is not as novel as it seems, in that he is just restating what Marx saw as the fusion of technology and economics, refracted through Deleuze&Guattari[8].

- Where Land might depart from Marx, is when seems to ascribe a sort of *agency* (or in its most provocative framing, a self-organising planetary "intelligence"[9]) to capital that feels heterodox at first glance. However, Land emphasises that his view (again) is the price system as one of distributed computation and resource allocation, something that is essentially restating Hayek's insights and reframing them for an age of networked, and perhaps, artificial intelligence [TU 2024].[10]
- Land has long been obsessed with networked technology (such as the internet but also blockchain and distributed cryptographic techniques that post-date the canonical CCRU period), and of course AI, particularly in the possibilities of feedback loops and recursion that entered a broader cultural consciousness with the internet of the 1990s. He cites Wiener and Shannon, in particular to criticse the tendency in mainstream thinking on computational systems and information theory for homeostatic, self-

[6]Cite this as teh 'conventional' definition of capital.

[7]From his 2018 essay on Bitcoin: "sequilibrium, "governed"—or, more strictly, made radically ungovernable—by a fundamental positive-feedback dynamic."

[8]see this 2017 post for Land's definition of (de-)territorialisation and positive feedback loops as inherently nihilistic, and at least in Marx's view (albeit the latter was writing in the context of free trade rather than modern networked technology), the only way forward to achieve revolutionary social change. See also the chapter on Marx's *Fragment on the Machines*, as well as the contributions of Camatte, and Deleuze-and-Guattari in [Mackay 2014]

[9]Land, in a recent podcast, defined intelligence as "competence in playing games, with a very wide definition of games that encompasses most activity" TU 2024

[10]Find bit of TU where he says this, and ideally somewhere in the texts, and connect to Hayekian

moderating or negative feedback systems. Land seems to be more interested in the possibilities of positive feedback, which (again in an echo of Marx) is something he sees as possible or inevitable in systems of capital.[11]

- It isn't clear how deeply he engages with recursion as it applies in AI, e.g. self-referential systems, Church-Turing, incompleteness, etc., however his discussions of Gödel's contribution around diagonalisation might inform Land's views on numerology and cryptography (which perhaps presages his writing on blockchain), levels of abstraction in computational systems ([Land 2011] pp. 43-44).[12]
- Land's interest in looping structures finds its most tricky, from a common-sense (not to mention rationalist) perspective, expression when applied to time. He sees one of capitalism's tendencies as being a compression of technological development cycles[13] He also talks about a type of temporal recursion, where future intelligences (including superintelligence) actively shape their own development backwards in time through the mechanisms of the market. It isn't immediately clear whether he means this in an acausal and literal sense, or if he is looking at it, in some sense, "outside of time".[14] As he writes: "This is because what appears to humanity as the history of capitalism is an invasion from the future by an artificial intelligence space that must assemble itself entirely from its enemy's resources." ([Land 2011], p. 338), a phrase that has a hint of the Gnostic distaste of the [demiurge for its creation].[15]
- Land's writing also has a significant dose of the occult, which I won't consider here, but

[11] A little footnote with some gloss on Wiener's ideas, Shannon, homeostasis from Lem, Maturana => note that this will come up in future chapter on S-o-M

[12] Not sure how relevant this is, but see Parisi in Accelerate for a short intro to Godel/Hilbert/Turing/Chaitin and what the broader context is. "Diagonal" is also mentioned twice in Swarmachines.

[13] Source/significance?

[14] Find source, podcast has stuff 1:53:00

[15] Note p. 624, PKD reference, doesn't really make much sense

which might form an infrastructure to at least some of his AI and capitalism related thought.

Markets as computational systems

In its most interesting, and least literal version, Land's thesis is that the market is a computational system that has features of something that looks like agency; moreover this distributed computation *is* a proto-Artificial Intelligence[16], and that the internet and the various stages through which machine computation has evolved so far, can be seen as (or perhaps actually are) teleological stages in the self-realisation and recursive improvement of some (super)intelligence-to-come, as it bootstraps itself into existence using the tools (e.g. intellectual and physical efforts, capital, organisation) of its human creators. That is, humans are just some cognitive substrate (generated by another impersonal process, namely evolution) that happens to run on wetware (neurons powered by metabolic processes) building machines (that currently run on silicon powered by a mix of energy sources that ultimately trace back to the low entropy energy of the sun or to the nuclear binding energy of the atom).[17]

Antecedents: the Austrian school

Part of Land's conceptual grounding for his view of markets as a type of distributed intelligence comes from economists (in one branch) of the "Austrian School", particularly

[16]He seems to mean "Artificial Intelligence", in the sense that there is computation happening that is increasingly good at matching human capacities. He also (in slightly different contexts) refers to AI systems meeting various Turing-like tests to greater degrees [TU 2024].

[17]What is the technical term for bullshit this might sound like? Teleological? Some fallacy?

Friedrich von Hayek.[18] These economists saw market economies as information processing systems that were distributed across many actors (people and firms), which often behaved irrationally and on imperfect information (much of it tacit knowledge as theorised by Michael Polanyi), but somehow their aggregate activity allowed for efficient prices to be set.

Hayek (and his collaborator Ludwig von Mises) was making his arguments in part as a counterpoint to the "socialist calculation problem", that is, the difficulty of finding the clearing price (i.e. the relative prices at which goods or services are efficiently allocated) for a given good/service in a complex economy, in a centralised way, because the preferences and constraints of various actors are distributed, are essentially "in their heads". Hayek's specific take was not just one of practicality (how is the central planner to find out these preferences), but epistemological in that these preferences, being tacit in many cases, are inherently incommunicable.[19]

Land seems also to have incorporated ideas from Eugen von Böhm-Bawerk, who developed a more complex theory around types of capital stock that exist at multiple removes from the actual consumption object (e.g. the fishing net, or fishing boat, that is used to catch fish; at a second-order, the machines that produce these nets and boats; these innovations increase time and cost associated with fishing, but potentially increase productivity, and depending on one's view, might be emblematic of a tendency of capital to enable complexity).[20]

[18] Cite at TU 2024 ~30:00, point out that neither Accelerate nor Fanged Noum actually mention Hayek, so is this lineage backfilled?

[19] Would be good to find sources and get a LLM to check this; also mention Popper?

[20] Land's actual written references-by-name to Austrian school economists are thin, other than this quote in *Teleoplexy* [Accelerate 2014]: " 'Acceleration' as it is used here describes the time-structure of capital accumulation. It thus references the 'roundaboutness' founding Bohm-Bawerk's model of capitalization, in which saving and technicity are integrated within a single social process-diversion of resources from immediate consumption into the enhancement of productive apparatus. Consequently, as basic co-components of capital, technology and economics have only a limited, formal distinctiveness under historical conditions of ignited capital escalation." However, he explicitly credits Hayek *et al* in [TU 2024], and [Fitchett 2020] digs more into the Hayek connection.

In a curious aside, the calculation problem in economies, far from being a pure academic dispute, was something that had concrete expression in the Cybersyn project of Stafford Beer (in Chile) and the rather different OGAS experiment in the USSR, championed by Viktor Glushkov. There is of course a broader field of study on the extent to which the ideas around Hayek, Mises, Popper, and others were deeply affected by both the aftermath of World War II and the Cold War's Manichean fight between capitalist (both US-style and social democratic) and Communist systems; however, this falls out of the scope of this project.[21] In any event, Cybersyn and OGAS both failed, for different reasons, but a live question today is whether highly networked societies (internet but also on-ledger transactions) with the power to run large-scale simulations might be able to find clearing prices, where earlier attempts failed.[22]

Austrians and Austrians The Hayekian branch of Austrians emphasised decentralised price-discovery and setting mechanisms, and as such, had a basically opposite orientation to the rational agency and game-theoretic origins of modern conceptual thinking about AGI (e.g. the MIRI View, discussed below). Recall that expected utility maximisation (under Von Neumann-Morgenstern) had 4 requirements: agents must have clear, consistent preferences, that can be ranked, the probabilities of which are known, and the agents have the computational ability to assess their choices.[23] Hayek's views, on the other hand, were founded upon information being practical or tacit, not being available centrally or in common, high levels of uncertainty, and agents that were often acting irrationally. Notwithstanding, Morgenstern, whose origins were with Mises, eventually (after meeting von Neumann) shed his

[21]Maybe Medina's book? Check *Collapse*?
[22]Find sources and what is Land's view on this - seems to touch on it around TU 2024 30:00?
[23]Is this discussed in omohundro? Probably should be?

early reticence about mathematical methods to find equilibrium prices (something Hayek continued to be skeptical of), and embraced what would be known as game theory, as a way to model situations with multiple agents, at least for certain classes of problems.[24]

Inevitability

Land's teleology has an air of inevitability to it: "It makes no more sense to try to rescue the economy from capital by demarketization than it does to liberate teh proletarian from false consciousness by decortication. In neither case would one be left with anything except a radically dysfunctional wreck, terminally shut-down hardware." ([Land 2011], p. 340)[25]

Substrate independence

This "substrate independence" is a different reading from Bostrom's version[26] and sounds weirder, but when added to one of Land's more controversial points - that humans are, via technology, creating the conditions that will eliminate their own evolutionary niche[27] a view also found in Hendrycks 2023 and Metzinger 2017 on humanity's "ugly biological bootstrap phase".[28]

Distributed systems win

Land's Hayekian intellectual lineage (though he vigorously resists that term in [TU 2024]) with its overtones of distribution information gathering and processing, seems to echo his

[24] Substantiate

[25] Figure out a way of referencing critical people who say AI can be shut down - from pause/remmelt to Ruha Benjamin vs those who say it can't be stopped owing to coordination problem, and once built cannot be contained owing to boxing problem

[26] Define/cite

[27] Cite in TU 2024 ~15:00 but also in writing

[28] What about Heidegger - does he say anything about this?

interest in distributed cryptographic systems (i.e. Bitcoin and blockchain). However, he explicitly points out the decentralised, distributed, non-planned nature of current machine learning systems, which "construct" algorithms via SGD (and such algorithms as we can, at present, not recover or write down as a conventional program), as the instantiation of his philosophical perspective in the teleological self-creation of AGI. ([TU 2024])

Divergence from transhumanism

In the sections below, it will be clear that some significant part of alignment's origins comes from the tranhumanist and futurist movements (e.g. Bostrom and Yudkowsky). Transhumanism, for many people, has an "ick" factor that perhaps Land of the alt-right years also evokes.[29] However, it is worth noting and relevant for our purposes, that they are not the same. Bostrom and Yudkowsky, in their writing, very clearly hold a special place for humanity, our values (insofar as they can be determined and aggregated), and a future that includes our descendants, whether or not they are recongisably biological.[30]

Most difficult to accept from multiple perspectives (left-coded, as well as centrist or moderate right-coded, as well as the mainstream AI safety/alignment view that is described in this chapter), Land sees no "special" status for humans or what they do, and thinks of humanist commitments as little more than a species level "existence bias"[31] or simple preference for that which is similar.[32] Land whether in *Circuitries* ([Land 2011], pp. 293-

[29]At the risk of digressing into the meaning of ill-scoped [suitcase perjoratives], the neologism TESCREAL is used by some to gesture broadly towards the aspects of AI that they find objectionable, but it isn't clear (yet) whether Land's thinking has a specific place withing Gebru/Torres' [pan-devil's house neologism like pantheon?]

[30]This itself is a major question, a locus of changing opinion amongst writers including Yudkowsky, Christiano, etc., and a central question of this project. [Cite sources for this.]

[31]Cite Metzinger and explain that context

[32]Hint at Carlsmith and copious writing trying to avoid this problem or at least make it more robust and clear, which is topic of rest of the diss.

294), or as recently as 2024, never really gets off the "train to crazy town"[33] and sees the future as exclusively a teleological process defined by capital and intelligence, interacting with humans who are, at best, historically necessary but prospectively marginal players ("antechambers to experimentation, cramped and parochial places to be"). He describes a future with a positive dissolution of culture into a "planetary technosentience reservoir".

Imagined dialogue between Land and an alignment researcher

Readers of this section might benefit from an understanding of current issues in the alignment of AI systems to human survival as a biological species.[34]

There are a numbers of responses that a given alignment researcher[35]

I've used a combination of models to guide this debate[36], but the main takeaways are as follows:

- Land speaks of superintelligence in a quite reductive way (in the sense that Bostrom or earlier writers might have done) without considering the detailed arguments set out by various writers on why even near-current level AIs could have severely damaging impacts, or without considering how current or proto-AGI systems might be structured (in terms of forming goals, planning, maximising reward, etc.), points that alignment researchers critique about the MIRI View (for instance).

[33] Locate this phrase, and check the TU 2024 podcast to see if it's true.
[34] Maybe should be moved to alignment chapter
[35] Say, one that was a compressed version of Paul Christiano, Eliezer Yudkowsky, Matthew Barnett, Katja Grace, Ajeya Cotra, Evan Hubinger, Daniel Kokotajlo, janus, Buck Shlegeris, Rohin Shah, Beth Barnes, Joscha Bach, Andrew Critch. This list is subjective, but I have used the following criteria (based on alignmentforum.org rankings): personal familiarity with views; gender balance (4.5/13 are not, to my knowledge, male-identifying); a mix of technical and philosophical perspectives; volume of output and community scoring/commenting.
[36] Attach transcripts

- For instance, his definition of intelligence as "game-playing competence" [TU 2024], is, at least, inadequately specified (what exactly is a "game"?) - which feels vaguely unsatisfactory for such a load-bearing term.[37] In the podcast, he seems to define the term in maximal terms, rendering it potentially unarguable or unanalysable.[38]
    * Some other (alignment-discourse relevant) conceptions of intelligence (from Claude-3.5-sonnet-new) include: optimisation power (the ability to hit a small target in a large search space); intelligence as building/updating world-models that minimise predictive error; ability to generate abstract patterns and manipulate them (not necessarily with any goal) and ability to transfer abstractions across domains; ability to learn and satisfy complex preferences that don't easily correspond to concrete objectives; ability to reason about counterfactual worlds; ability to integrate information across scales and modalities; ability to explain phenomena through minimal descriptions (e.g. compression); ability to meta-learn (learn how to learn, which is more about process than outcome); ability to manipulate causal models[39]
- Land does not claim to engage with developments in deep learning, RL, or even LLMs, and his only references to the AI literature seem to be people like early, e.g. I.J. Good, the fiction of Samuel Butler, mathematicians like Gödel and Hilbert, or of course philosophers. This is a problem in the sense that AI (circa 2024) is an empirical discipline, more so than it was in 2011-2014.

[37] I can't find it explicitly defined in [Land 2011] or [Mackay 2014], though Reza Negarestani in the latter volume, does take up the topic in his essay *The Labor of the Inhuman*.

[38] Get the quote

[39] Look in sonnet transcript for names, might need further explication/references perhaps from LW/PIBBSS reading

    * For instance, researchers seem to be confused as to how human-given goals are actually represented within a model (the inner alignment or mesa-optimisation problem); the problem of "what does the model really believe?" (the Simulators framing); are all AIs necessarily goal-directed or agentic (again, the Simulators framing and contemporary work on scaffolded agents); might near-current models learn human (some approximation of some subset of) values through RLHF, so that alignment actually is easier than originally supposed (in the MIRI View)

- Insofar as markets are doing anything describable as optimisation, they seem to be doing this locally, rather than at any global or broad scale, and there doesn't seem to be any goal or overall target towards what markets are doing (any more than evolution, which seems to operate towards a very local purpose, namely genetic fitness across a single generation, isn't normally thought of as having a trans-generational goal or optimisation target)
- Land's teleological approach seems to imply that human actors (e.g. in governments, regulators, corporations, as well as scientists) have no substantial ability to influence the course of the near future, and the forces of runaway market-based optimisation would lead to scenarios that disempower humans. This has some apparent similarity to the writing around "Moloch" and general international and corporate coordination failures, however Land is going further by seeming to impart agency to the system itself.
  - in response, GPT-4o (acting in the persona of Land) says:

“[My] argument draws from a tradition of thought that recognizes complex systems—including markets—as exhibiting emergent behavior that can, over time, appear to operate with a kind of teleological direction. This is not a mystical or supernatural claim, but a recognition that certain systemic dynamics have a momentum of their own, pushing towards outcomes that transcend the intentions of individual human agents. . . .This perspective is rooted in a deeper understanding of systems theory, where distributed networks—whether in economics, biology, or computation—can generate behavior that appears purposeful when viewed from a certain scale. Hayek’s notion of the price system as a form of distributed computation is a case in point: no single individual controls the price system, yet it allocates resources in a way that appears coherent and, to some, “intelligent.”. . . Moreover, my argument is not that markets are actively aiming to produce an AI that eliminates humans, but that the dynamics inherent in our socio-economic system are conducive to such an outcome if left unchecked. Capital’s inherent drive towards self-multiplication and acceleration—what Marx saw as the tendency for the rate of surplus value to rise—leads to a relentless pursuit of optimization and efficiency, with technology as the primary vehicle. . . Human actors, companies, and governments certainly make decisions, but their choices are often deeply influenced by the structures in which they operate. Attempting to halt AI development is not merely a matter of individual or collective willpower; it involves grappling with the underlying incentives, pressures, and feedback loops that drive technological and economic systems forward. . . Markets, as

> expressions of [Delezo-Guattarian deterritorialisation] process, can "slip the leash" of human intentions and generate consequences that no single agent, nor even coalition of agents, intended or foresaw…I am not denying human agency, but suggesting that it operates within constraints that are more like currents in a river than the blank slate of a field. One can paddle upstream, but the flow exerts a pressure that shapes—and sometimes overrides—our choices."

Contents

What is this chapter about

This chapter covers the following topics:

- sets out why AI as a technology is "special"
- discuss the difference between AI alignment, safety, and ethics
- explain why AI alignment is of particular interest
- the contingent history of AI alignment as a discourse
  - examining the judeo-christian allegation
    - what were the imaginaries that served to communicate ideas?
  - the transhuman background
  - what is the relevance of x-risk (and potentially, s-risk)
  - how has the AI x-risk threat model changed
- the MIRI view, impersonal optimisers, recursive self-improvement

- what futures have value
- how did technology shape discourse
  - GOFAI
  - RL
  - next-token predictor
- what were the internal critiques of MIRI view
- revisiting the external critiques
- is alignment a relevant concept or discourse today
- how do we look at the above topics (which are AI and AI alignment-specific) against the positions articulated in the CCRU, particularly by Nick Land

Contextualising against the project as a whole?

This chapter's central claim is that 'alignment', taken less as a literal term and more as a discourse[40], had certain contingent origins that continue to affect the priorities of the field and researchers in it, may affect their world-views, and influence surrounding contemporary conversations. The chapter also draws a parallel, through anecdote and enriched through practice, with accelerationist thought (of the Warwick CCRU variety) and a fertile, perennial theological-apocalyptic vein in Western conceptions of technology. While possibly more coincidental than causal, these parallels and analogies inform the project's practice component.

[40] I'm using the word 'discourse' in the sense of Paul Edwards' book *The Closed World: Computers and the Politics of Discourse in Cold War America* (1997) (..**quotation**).

Historical Bases of Alignment-as-discourse

The claim of this chapter is that AI alignment, both as word and discourse, became prominent in the years between 2008 and [2014] and came to incorporate certain ideas relevant to AI that were circulating pre-2008, ideas that had various antecedents, including technical ones such as microeconomics-based theories of rational agency, operations research, and computability theory. These technical frames of the problem of AI existed (or at least were sometimes communicated) in a milieu that had imaginaries drawn from (science) fiction; at the same time, these frames were being used to actively battle a particular Hollywood-framing of AI as an anthropomorphic killer robot (Terminator).

This was a proselytising exercise, the principals of which came out of an earlier on-line milieu of transhumanist futurists. As of [2014], alignment was still a mix of (mostly) non-quantitative philosophical futurism (e.g. Bostrom, Sandberg), and mathematics-based explorations into the nature of agency (e.g. MIRI's agent foundations and decision theory agenda). Between [2014] and [2019], this largely theoretical work started coming into contact with reality as interesting results in reinforcement learning appeared. Alignment's definition (as practically expressed through discourse) was enriched, and the abstractions of the philosophical and mathematical sub-discourses were questioned, or increasingly, treated as fascinating but unverifiable. By [2019], the ground shifted again, as reinforcement learning gave way to language models as the most likely path to AGI. The concepts that grounded alignment needed to be revisited, and often, reworked or replaced to accommodate the drastically different architecture of LLMs. Currently, as of [2024], alignment has become a less popular term, as the field (of AI) has grown and powerful AI has become more tangible, terms that are more inclusive of actual or near-term and concrete risks, such as AI Safety and

AI Ethics, have become more common. The aspects of alignment that were most empirically supportable, as well as the most useful theoretical framings, were incorporated into the broader conversation, while other sub-discourses have petered out.

Why does anyone care? - specialness of AI as GPT, as amplifer (Russell's WEF talk) with in-built philosophical/social aspects - general purpose problem solvers as goal-directed planners

What is Alignment?

- Given it is a weird word;
- why are we interested in it ;
- what does it mean literally;
  - confusions on values vs don't-kill-everyone
- what were the features of the particular view I'm interested in here
  - self-replicating generally capable intelligence
  - power imbalance vis a vis collectives of humans
  - perfect rationality
  - people promoting this view fundamentally still had a human attachment (reza's critique I think)
- how common is it and what different ways is it used (quantitative)
  - connect to antikythera
- the friendly angle, safety and ethics show up, alignment gets repurposed
- who are the relevant people (make a glossary or exhibit):

- MIRI[41],
- EY, NB, Omohundro

A timeline would be really useful

Origins: Rational Agency

Sources

In launching this historical study, I focus on a few sources, which seem to be the earliest mentions of 'alignment' as an important topic around advanced AI systems.

Discussions around alignment are better appreciated in the broader context of (some of) the various disciplines that informed AI[42]:

- philosophy (from Aristotle and Descartes on humans as creatures whose actions are guided by reason, through Hume (on induction), Carnap (on the mind as a computational process), Wittgenstein, Russell and the Vienna Circle (on empiricism as the handmaiden to rationalism), Bernoulli, Bentham, and Mill (on utility as a way of capturing a human's subjective valuation of some outcome, which leads to utilitarianism when applied across populations), and Kant (on rules being an appropriate guide to action, rather than blindly following a utilitarian calculus))
- mathematics (Bayes (on how to update beliefs based on new evidence); Gödel; Church, and Turing (what problems are computable within any given system or architecture))

[41]The Machine Intelligence Research Institute was a major early incubator of thinking around agent foundations, which is the study of the theoretical foundations of agency, both in humans but also across societies. *Note from google doc, this is contested - check what agent foundations is and why it's important in this context - Alignment borrowing from agent foundations without justification isn't obvious. David Mannheim's paper 'reasons for working on highly reliable agent design'*

[42]mostly taken from Russell & Norvig 4th ed Ch 1

- economics (Bernoulli on diminishing utilities (at least for money); Walras, von Neumann, and Morgenstern (who broaden ideas around utilities and preferences-for-outcomes to contexts broader than monetary or economic ones); the latter two also do formative work in the theory of games)
- operations research (Bellman on Markov processes which govern situations where payoffs are delayed in time and depend on a sequence of actions; Simon on satisficing)

Russell/Norvig describe the other disciplines (cybernetics, computer science, psychology, linguistics, neuroscience) in much greater detail, but for purposes of this chapter, I set this truncated background.

Omohundro in 2008

Self-improving systems ~ rational agents The earliest mention of 'alignment' appears to be Omohundro 2008 on self-improving systems, by which he means humans and other naturally-evolved systems (to some extent) but also human-designed computers that can change their software and hardware in order to achieve goals in some environment.[43]

The abstract of this paper is instructive:

- self-improving systems as a coherent concept
- the idea that self-improving systems converge on certain architectures, and that they can be understood through certain tools originally developed in a microeconomics

[43] Claude's analysis was only 3 citations in the paper's bibliography were alignment related (1, 2, 22). There were 15 other citations in computer science, 8 in physics/astrophysics, 9 in economics, probability, and game theory, 5 in evolutionary biology/psychology, 4 in human psychology/cognitive science/linguistics. The bibliography had 48 entries.

context

- self-improving systems tend to develop four 'drives' (towards efficiency, self-preservation, resource acquisition, and creativity)

Taking these in turn: the idea of self-improving systems (as the highest level of a hierarchy that goes progresses in complexity, environmental responsiveness, and adaptiveness through inert, rigidly reactive (thermostats), adaptive (basic physiological homeostatic systems), deliberative (human reasoning), culminating at self-improvement (which humans can only do in very limited ways)) is documented within computer science at least to 1965 (I.J. Good)[44].

An important observation is that self-improving systems, far from being unpredictable, seem to be amenable to microeconomic frameworks such as that of von Neumann and Oskar Morgenstern (1944), Leonard Savage (1954), F.J. Anscombe and R.J. Aumann (1963)[45]

Omohundro's quote on self-improving systems is instructive 'Self-improving systems do not yet exist but we can predict how they might play chess. Initially, the rules of chess and the goal of becoming a good player would be supplied to the system in a formal language such as first- order predicate logic1. Using simple theorem proving, the system would try to achieve the specified goal by simulating games and studying them for regularities. By observing its patterns of resource consumption, it would redesign its chess board encoding and optimize its simulation code. As it discovered regularities, it would build a chess knowledge base. General knowledge about search algorithms would quickly lead it to the kind of search used by Deep Blue. As its knowledge grew, it would begin doing "meta-search", looking for theorems to prove about the game and discovering useful concepts such as "forking".'[46] He

[44] ref p.3 omohundro 2008

[45] Omohundro does point out that this conception of *homo economicus* as a rational actor is increasingly being challenged in the case of humans, e.g. in behavioural economics.

[46] p. 6

goes on to describe how these systems might (if they have the affordances to do so) might redesign their own hardware in the service of efficiency.

He goes on to formalise the intuition that rational economic actors have internal representations of goals; identify the possible actions in a given environment and assess the consequences of each action; pick the action that will achieve the goal with high probability; observe the environment and then update internal representations appropriately.[47]

He claims that a certain microeconomic formalism should drive the behaviour of rational beings; the presents the concept of 'vulnerabilities', conditions that cause an agent or system to lose resources (money, in an economic context, but this could be food or sleep, and more generally, physical things like free energy or useful matter) created by the interaction of the agent's environment and its particular preferences. Sometimes these vulnerabilities are actively exploited by (other actors in the) environment, and in other cases (like the bird that repeatedly flew into Omohundro's shiny car fender to the detriment of its genetic fitness) it is evolution in some sense that needs to 'work around' such vulnerability.

If there are such vulnerabilities, the agent faces a pressure to optimise, and the resulting system should have certain features: beliefs (what an agent knows about the world); using probability distributions to represent beliefs; and the use of utilities as a quantitative statement of preferences; particularly in the case of self-improving systems, the importance of separating what one knows about the world (beliefs) from what one 'wants' (preferences); and discounting outcomes in the future, potentially at a different rate depending on how far in the future they are. The latter in particular might lead to issues around infinities, as well as over-weighting the present and under-weighting the long-term future[48].

[47] p. 7

[48] topic that we return to below in the discussion around existential risk and longtermism

Another key insight of the paper (drawing on literature on algorithms and computational efficiency) is that although future AIs may be powerful (compared to humans or present AIs), they are very likely to face resource constraints on how much computation they can undertake - the real world is far more complex than anything humans or evolution are likely to create. Agents are always operating with insufficient resources, and must learn to economise and allocate in some way. Omohundro describes the notion of 'proxies', which can be of various sorts, but basically allow computations to complete faster or more cheaply or not happen at all. But this reduction in complexity often comes at a cost (the map is not the territory).

Rational agents have certain drives Omohundro's paper is better known for its notion of "drives", that in his view fall out of this framing of rational agency. If one accepts Omohundro's starting premise (that self-improving systems tend towards being rational actors in a microeconomic sense), then his assertions about the drives (efficiency, self-preservation, acquisiton, creativity) becomes easier to see. Yet even here there are subtleties that will continue to show up, perhaps elaborately masked, in subsequent discussions around alignment.

His writing on efficiency is best-grounded in that it mathematically carries forward the concept of utility to describe how a system might allocate resources amongst its sub-systems; what the physical (communication latency and energy consumption) limits on a computational system might be; the deep role of entropy in computation. The most intriguing aspect is that he predicts that sufficiently advanced systems develop manufacturing techniques that are "atomically precise" (in a specific entropy-preserving sense), which might inform Omohundro's subsequent work on nanotechnology. This is potentially interesting, at least in hindsight, since the “AI x-risk imaginary” envisions the universe turned into “grey goo”.[49]

Another section on self-preservation considers the need for a rational agent to hold on to its utility function, as this is the central core of “who they are”, thus butting up on personal identity. Specifically, if future AIs are able to create arbitrary numbers of backups, their sense of identity may or may not be as bound up with physical instantiation, and they may instead seek to preserve their utility functions (assuming these encode their most important goals). In an echo, albeit uncited, of philosophical thought on non-personal identity (Parfit 1998), identity in a Buddhist context, or open individualism[50], he states that certain systems might sacrifice a given physical instance if “it could be convinced that another system was as dedicated and could use its resources more effectively for this cause, however, it might willingly sacrifice itself.” (p. 26) In another rhyme with Land’s thought, the self-preservation tendency might mean that war itself changes - there is no point destroying the enemy’s computers, rather one should want to scramble its cognition - all war becomes informational.[51]

Omohundro’s work, and much of the subsequent alignment discourse as well as [Lem of the summa], relies on evolutionary analogies. For instance, in part 6 of his paper, on the drive towards resource acquisition, Omohundro refers to [Smith/Szathmary] on the “hard steps” in evolution, an idea which shows up in the literatures around alignment, the Fermi Paradox, and the shape of alien cognitions - topics that deeply inform the ethical and philsophical motivations behind AI alignment and existential risk reduction.[52] Whether one is using

[49] See if any of grey goo cites this paper or nanotech. Any other names for this?

[50] Get some sources, this is kinda important around corrigibility question, stuff with nick dupuis, and feeds into bostrom cosmic host/VFN. Perhaps these ideas can be brought closer to ECL/decision theory.

[51] Perfect place for a Land quote

[52] This is kind of important - it isn’t just alignment as a computer science problem…the basic motivation for all this (or at least the claim of the thesis) is that that unlike previous technolgies, self-improving optimising processes have the potential to bring about an ‘end without remainder’ to human existence, and given that we

the evolution or astrobiology framings of evolutionary hard steps, some convergences are: creation of self-replicating molecules, the transition from prokaryotes to eukaryotes and multicellular organisms, and then to colony-based organisms with culture. Omohundro's invocation of Smith/Szathmary smuggles in another assumption, or perhaps a desideratum, of advanced societies - that they necessarily develop cooperative and prosocial behaviours as a way retaining cohesion and longevity.

This correlation of cooperation and societal complexity is exemplified by Omohundro by multi-cellular organisms' immune systems as defences against cancer (reading cancerous cells as an antisocial entity that exploits the surrounding environment), and citing a documentary ("The Corporation", 2003) corporations as quasi-sociopathic entities (p. 29), that amass resources albeit within regulatory constraints.[53] In the context of self-improving systems, Omohundro sees free energy as the most important constraint (where "free energy" is that energy which is useful for work, since some portion of kinetic, potential, thermal, or chemical energy is inevitably lost in entropy increases, at least in common Earthly situations, and bracketing the specific entropy-preserving atomic scale processes Omohundro proposes). Omohundro briefly summarises the importance of solar (and stellar) energy as the purest (e.g. lowest entropy) energy available; less efficient but plentiful energy is also accessible through nuclear fission and fusion. The importance of stellar energy implies that resource-hungry systems should expand into space, perhaps through Von Neumann Probes,

haven't found anything else in the universe after ~60 years of searching, and have some theoretical models to say we might not, whatever we are doing here is in some sense important, if only for its rarity. This relates to hanson's great filter paper from 2008 or bostrom's urn. **This should be brought out at a higher level, if nothing else - humanity isn't stumbling into the abyss, there is quite a lot of writing from x-risk and accelerationism pointing the way.**

[53]This is another important seam - that companies/corporations are sociopathic - shows up in Christiano - trace this line through to letter to future, which takes the more market-friendly view. This also aligns well with Land and perhaps others in ccru

developing structures like Dyson Spheres to capture the maximum energy from a star, and perhaps ultimately restructing at a cosmic scale.[54]

Is/Ought Distinctions Importantly, Omohundro's framework is not saying anything about how 'good' (in whatever sense) any goals actually are - they may be selfish or altruistic, or anything else, as long as the goals can be mapped to utilities. Although his analysis is largely descriptive (of how physics-constrained systems achieve certain design constraints), a discordant normative note comes from his discussion of creativity (one of the drives) where he seems to find a maximally efficient universe somehow unsatisfying: "Once it had satisfied all the material goals, this kind of limited agent would have no greater purpose. With too limited a vision, the universe might become an efficient but distinctly empty and non-human place." (p. 31) He goes on to explain much of human cultural production as a "costly signalling" behaviour ultimately grounded in prospective reproductive success, citing the evolutionary psychology literature. He suggests that we (in designing future AI systems) should be careful not to lose such costly signally behaviours or the resulting artefacts, which would be the (by his own logic) tendency of self-improving systems, which "will be motivated to find ways to make [such signals] without the cost. For example, a system might demonstrate its intentions directly by displaying portions of its utility function. If we want to retain the richness generated by costly signalling, then we must find ways to keep it from being eliminated by improvements in efficiency." (p. 32)

This "ought"[55], in a work that is otherwise relatively descriptive, seems to be consistent

[54] Ref to cirkovic, collapse volume on this, bostrom. maybe talk to mackay. The kardashev concept should be mentioned - maybe a claim of this thesis is that humanity has a destiny. **alignment is not just a cautionary project (of not going extinct), it is also promethean (realising our destiny)** Show the seam into anders sandberg etc.

[55] Ref is/ought

feature of the discourse around AGI - the potential consequences of the technology are described, but the relevant authors seem to struggle with the implication (human extinction or the extinguishing of costly signalling behaviours or a universe with no art/music/meaning).[56] Land repeatedly notes (e.g. in [TU 2024],) that he is merely describing how he sees the world, rather than prescribing how it ought to be. In this sense, Land's accelerationism is more clear - it bites the bullet of its reasoning, in that Land recognises that the combined historical forces of capital and AI mean the end of biological humanity as we know it.

The MIRI View Omohundro's paper set out some of the key preoccupations that would continue to inform the alignment discourse, but another principal source of basic thought about 'the AI problem' came from the Machine Intelligence Research Institute[57], set up by Eliezer Yudkowsky, who perhaps did the most (at least in the 1990s) to make the idea of out-of-control AI "tangible", at least for a mathematically or scientifically literate audience.[58]

MIRI's archive of research starts with a publication by Yudkowsky from 2001, that sets out his original vision of creating benevolent or "friendly" AI, which is a term less-used now, but is a seemingly quotidian term that is used by Yudkowsky in a multivalent sense.[59] In section 2, he writes: "The term "Friendly AI" refers to the production of human-benefiting, non-human- harming actions in Artificial Intelligence systems that have advanced to the point of making real-world plans in pursuit of goals…Because of self-improvement, recursive self-enhancement, the ability to add hardware computing power, the faster clock speed of transistors relative to neurons, and other reasons, it is possible that AIs will improve enormously

[56] Make this a lot more precise - which authors are we talking about? Is this a major point - in a sense it is the major claim of the thesis and connects well to Land?

[57] Point to mission statement of MIRI and 1 sentence summary, mention it is defunct

[58] Brief blurb on Eliezer, sequences, HPMOR, etc.

[59] According Yudkowsky's post to the Arbital site, which dates from 2015, "Friendly AI" was a term proposed by Bostrom.

past the human level, and very quickly by the standards ofhuman timescales...Friendly AI is constrained not to use solutions which rely on the AI having limited intelligence or believing false information, because, although such solutions might function very well in the short term, such solutions will fail utterly in the long term. Similarly, it is [safe] to assume that AIs cannot be forcibly constrained...Success in Friendly AI can have positive consequences that are arbitrarily large, de- pending on how powerful a Friendly AI is. Failure in Friendly AI has negative conse- quences that are also arbitrarily large. The farther into the future you look, the larger the consequences (both positive and negative) become. What is at stake in Friendly AI is, simply, the future of humanity."

Friendly AI seems to be a historically important term that continues to be in the MIRI archive[60] (and I extrapolate, the broader discourse) until about 2014, but was criticised as being excessively anthropomorphic and anthropocentric.

In 282 pages filled with his customary aphoristic text, often verging on verbosity, the document sets out his criticisms of anthropomorphic[61] thinking about AI leading us astray, as well as a box-and-arrow sketch of a possible AI architecture. The bibliography is one page, of twenty sources, six of which are science fiction; seven articles on AI; and five on evolutionary psychology, psychology, and cognitive science. There is also a glossary that is a fascinating window into the terms that continue to crop up in the discourse, almost a quarter-century later. The difference between this and the Omohundro document however, is the more quantitative grounding of the latter in game theory and economics.

The rest of the articles (total eleven) in the MIRI archive 2001-2009 deal with a variety of issues that are distinct from the Omohundro document in their emphasis on "big pictures"

[60] What about LW/Arbital/AF/sequences e.g. this seemingly good criticism from 2012
[61] weave in hayles' distinction between centric/morphic

questions around AI, and less so on architecture or nitty-gritty of building the things in a way that doesn't kill humans:

- how AI interacts with catastrophic or existential risks and geopolitics (three articles)
- how to load human ethical and moral intuitions (insofar as that is appropriate and can be identified consistently) and judgement into AIs (three articles)
- one article on mathematical details of utility functions
- an 122-page document on intelligence by Yudkowsky, viewed through frames or subsystems, that operate at different levels of abstraction (e.g. code, sensory modalities, concepts, thoughts, deliberation). Yudkowsky also considers how this framework and its implications might vary across minds-in-general[62]

2010 sees ten articles, at least one of which directly comments upon Omohundro's work (Shulman 2010), specifically considering cases where an agent following an expected utility framework might, depending on the "shape" of its utility function (e.g. whether it is satiated by large quantities of any given good, like humans often are for food, and perhaps even for money), opt to cooperate with other agents (including humans). It is hard to assess whether this analysis was impactful in subsequent discussions.[63]

Outside of [Shulman 2010], MIRI's archive 2010-2014 considers a range of topics including[64]:

[62]Do a deep read of this for minds-in-general chapter, and reference substantially in Aachen paper

[63]Maybe use notebookLM (docs are already loaded there) to search for references, and do same for lesswrong/stampy.

[64]This is TBD (**at least until I've assessed the 2014/2017 versiosn of MIRI tech agenda which presumably compresses the most valuable of this stuff**), but for the moment, I'm not digging into each of these - this diss isn't about MIRI's history and lot of these things become a background-of-sorts to alignment now, rather than mainstream research. This is just based on the titles - I tried to scrape all the relevant PDFs and then feed them to notebooklm (that project is saved in notebooklm but the actual subset of files that i could automatically download are in desktop/pystuff/scraper/miri_pubs while the titles that couldnt' be downloaded are in the smol/ dir). It might however be worth going into the various categories as they show up (outside of MIRI) subsequently under different names

- highly theoretical/mathematical thinking about issues that might apply to powerful AI systems unconstrained by anything other than physics (called "Highly Reliable Agent Design" in MIRI's parlance)
  - the various decision theories (primarily acausal) that had been developed (this will be the topic of subsequent chapters in this dissertation);
  - the mathematical and logical foundations of agency
  - how AI systems are to reflect upon their own reasoning (as humans do)
  - what is intelligence and what are its limits
  - possible Gödel problems with self-replicating ("tiling") agents
- problems surrounding value and values ("Value Specification")
  - how to transmit human values (howsoever defined) to an AI
  - how to ensure that AI systems deal adequately with uncertainty, which comes in various flavours (e.g. uncertainty about the human overseer's "intentions" versus what they actually said; empirical uncertainty about verifiable facts; uncertainty about normative things, such as "what should one value?");
  - the possibility of, and importance of avoiding, unintended consequences
  - how to ensure an AI that creates subsequent AI systems is able to preserve and transmit the original (human-derived) values
- the trajectories of AI development ("Forecasting")
  - paths to existential or catastrophic level risks[65]
  - the economics of societies with powerful AI systems

[65] Need a footnote explaining the difference and ref'ing Macaskill, Bostrom, SJ Beard, the new 2024 book.

  - uploading of human brains into software systems as another route to enhancing the aggregate cognitive capacity available to humanity
  - how useful are analogies from evolution, and from humans
  - the science of prediction
  - surveys of AI practicioners on timelines

- how to deal with various that might crop up, mostly on the part of the AI (but via humans' inadequate design of said AI), called "Error Tolerance" by MIRI
  - how to ensure that AI's values can be changed by humans later (given that early AIs might be badly constructed)

2011 Singularity Summit A slightly interesting document is the report of the 2011 Singularity Summit, which dealt primarily with whole-brain emulation ("WBE", which is creating software copies of human brains), in terms of the technical and conceptual issues involved, as well as how the timeline of WBE interacts with the timeline for AGI. The report primarily discussed how WBE (of AGI researchers or other scientists) could be a cognitive multiplier (100s of researchers working in parallel might, for certain problems, achieve results faster). The report (pp. 4-6) also had a number of interesting foundational questions that researchers were confused about, and it indicated probabilities of WBE arriving before various forms of A(G)I arrived.

The report was also interesting for its crisp statement of how an AI arms race would progress: "[Anna Salomon (co-organiser of the Summit) and researcher Carl Shulman, in a 2010 paper] argued that if whole brain emulation (WBE) came before artificial general intelligence (AGI), this might help humanity to navigate the arrival of AGI with better odds of a positive outcome. The mechanism for this would be that safety-conscious human researchers could be 'uploaded' onto computing hardware and, as uploads, solve problems relating to AGI safety before the arrival of AGI. Shulman and Salamon summarize their position this way: 'Business as usual' is not a stable state. We will progress from business as usual to one of four states that are stable: stable totalitarianism, controlled intelligence explosion, uncontrolled intelligence explosion, or human extinction. . . .The question, then, is 'whether the unstable state of whole brain emulation makes us more or less likely to get to the [stable] corners that we [favor]. Is our shot better or worse with whole brain emulation?' " (p. 2)

For purposes of this chapter, this document highlighted an ongoing view, that seems currently to be playing out amongst corporations but also geopolitically (at least from the US perspective), that an "arms race" (or "race dynamic") for AGI would have a high probability of resulting in a catastrophic/extinction-level outcome for humanity.

MIRI Technical Agenda (2014/2017) MIRI produced a technical agenda in 2014, which was revised in 2017, and which I take as presenting a compressed version of the types of things MIRI (and by extension, a significant part of the AI alignment world, at least up to 2014) was thinking about.[66] The foundation of this agenda, once one has accepted that superintelligent systems are technically possible, is that (potentially unbounded) resource acquisition drive would be a likely thing such systems would have, regardless of what their specific architecture or programming was, because more resources seem robustly useful for almost any goal. Such resource acquisition could easily lead to situations that are inimical

[66] I focus on the 2014 version to preserve a type of chronological continuity to this narrative, which is probably fictional. The versions have substantial similarities, but the 2017

to human flourshing or survival.

Formal Approaches to Agent Reliability

Starting from that premise, the document identifies 3 main planks to MIRI's agenda: firstly, ensuring an AI reliably pursues the goals humans ask it to pursue. Secondly, how do we define these goals to avoid ambiguity or misinterpretation? And lastly, can we ensure this AI can be modified, corrected, is helpful to its human overseers, and *in extremis* can be shut down, since we are not confident that the first (or any) such AI system will perform according to humans' intentions. MIRI's agenda, explicitly set out, so perhaps in contrast to other conversations in the discourse circa 2014, was focused on theoretical (as opposed to empirical i.e. deep learning-based) approaches, which they thought was important because of the possibility that an AI system could "pretend" to be helpful, harmless, and honest (a phrase that would subsequently be used in the broader discourse around making AI systems safe for humans), while "hiding" its intentions until some time in the future, perhaps when it was strong enough to rebel or defect (the so-called "treacherous turn").

Such a formal theory of intelligence and agency is relatively difficult (vis a vis other problems in physics/mathematics and non-human biology, and more akin to problems in philosophy, psychology/cognitive science, or sociology) because the systems being studied are deeply *embedded* in their environment, much of which reacts in agentic ways to the system we are trying to understand. Specifically what does the problem of induction look like in a situation where the agent performing the induction observes, acts on, and is in turn acted upon, by its environment? Moreover, what sort of world model does such an agent develop, and what ontology[67] does it use to internally specify its goals, and how does this

[67] Do linguistic dive into ontology and define what it means in this context.

ontology translate into other ontologies in the broader system or environment. How does such a system handle the fact that it is uncertain about what consequences any particular plan, particularly as it inevitably is operating with limited resources (relative to the number of decisions and likely complexity of the world)?[68]

Vingean Reflection

Another broad area of enquiry in the MIRI agenda concerned the idea of "Vingean reflection"[69] essay.], which goes as follows: suppose that many paths to superintelligence do not involve humans actually creating a superintelligent system, rather humans create one or more AIs that design subsequent more powerful AIs, which eventually culminates (ideally when we want it to) in a superintelligence. In such a world, we need to a) ensure the first or earliest AIs are aligned or safe (to/for human interests), b) that they can reliably transmit such properties as might constitute such alignment/safety to their successor AIs. Besides the general engineering and scientific difficulty this presents, there seems to be basic issue: how does one reason about the cognition of something that is, by definition, more powerful than one? As MIRI puts it (p. 7): "Any agent reasoning about more intelligent successor agents must do so abstractly, without pre-computing all actions that the successor would take in every scenario. We refer to this kind of reasoning as Vingean reflection." MIRI's goal in this respect would be to find formal (i.e. mathematical or logic-based) approaches to verify that a given AI has certain alignment/safety properties; however, the document points out certain problems (stemming from the incompleteness/inconsistency of formal systems as well as paradoxes that arise in self-referential systems) with achieving high reliability within formal systems

[68] MIRI is talking specifically about 'logical uncertainty'...would be good to have a crisp distinction between this and factual, normative uncertainty and why this is important (if it is, for this document).

[69] Named for Vernor Vinge's Vinge 1993

stemming from Kurt *Gdel*'s work, which makes this a spectacularly challenging problem.[70] The conundrum, besides its theoretical interest, continues to pesent in contemporary AI agends such as "scalable oversight" and "weak-to-strong generalisation".[71]

Error Tolerance and Corrigibility

It is very likely that early AI systems, like the HAL 9000 of *2001: A Space Odyssey*, will be error prone, or will pursue goals that have been imperfectly loaded by the human designers through methods that are in some way undesirable. In such cases, it is important that the AI be capable of correction or shutdown. The MIRI agenda in discussing this refers to Omohundro (2008)'s point that sufficiently advanced and capable AI systems might resist, possibly strongly, attempts to change their goals or preferences (as an imperfect analogy, think of preferences as akin to their "identity", which many humans would resist forcibly alteration of).

Value Loading/Learning

The MIRI document also discusses the difficulty of specifying to an AI system what one wants, what one's "intentions" are, owing to the fact that intentions are context-dependent, and the language they are expressed in is a lossy compression of what is actually desired by the speaker. This compression is often ambiguous and unclear, and in a human context, involves rounds of clarification and supervision to ensure what was intended is sufficiently close to the task that was actually performed.[72] A way of thinking about this slightly outside

[70]This is sufficiently interesting that its worth digging into and pulling out the threads that persist to today. At very least, examiners might go after this, and anyways it's **fun and connected to art.**

[71]Expand and explain - nice segue into actual stuff happening now

[72]This problem exists with humans instructing humans, but typically we have more or less good psychological models of others in our species, and most actions carry relatively low, or localised cost of failure. In the cases where the costs are especially high, we have evolved ways of minimising the risk of mistakes through misinterpreted intentions. The intuition behind the fears above is more "The Sorcerer's Apprentice" or "King Midas" (Russell/Norvig, p. 33).

AI is from Wittgenstein's considerable analysis of [shared life-worlds that contextualise the literal content of our sentences].

There are a few insights here that persist in current discourse: the difficulty of specifying intentions and values, for any given human, let alone a collective or population. Secondly, this section highlighted the inductive approach that had already been (since Ng/Russell 2000) increasingly used in reinforcement learning contexts, that is: rather than specifying objectives or preferences directly, get the model to "learn" these by showing it lots of examples.[73], with the immediate problem of how such example data is to be generated in sufficient quantity and diversity to cover most or all of the things humans actually value. Thirdly, in an echo of points above, the value the AI system learns must be stable as and when the AI designs and deployes new, potentially more powerful, systems into potentially different environments. Lastly, MIRI points out the fact that humans, singly or collectively, are not particularly clear or agreed on what "their values" actually are, particularly when projecting into the future (when circumstances might be different, and they might have more knowledge or more time to think).

Superintelligence (2014) The MIRI View has been discussed at length, but in 2014, Nick Bostrom's *Superintelligence* was also published. It was and continues to be influential, even though some of its ideas have lost currency. This chapter won't discuss it in great detail, except to note that it is a much broader document than anything else written prior, combining concerns similar to those raised by Omohundro/MIRI (albeit not until chapters 7-9, and 12-13); with analysis of the dynamics of how AI saturates the econonmy and ideas about state-level and international cooperation and the dangers of technological races (work that

[73] Quick gloss of Ng/Russell or point to chapter

is now described as “AI governance”) in chapters 4, 11 and 14; with musings on the prospects for human work and life in an AI-dominated economy (chapters 1, 4, 15).

There are several oft-cited ideas relevant to alignment (the term itself isn’t used, instead Bostrom prefers the term “friendly” (e.g. Box 11, p. 197 Bostrom (2014))):

- “intelligence explosion”
- the four categories of “oracles, genies, sovereigns, and tools”
- the spectrum of polarity: a global “singleton” AI vs multiple AIs in many companies/countries
- the Orthogonality Thesis

Although Bostrom put these, and many other, ideas together in one place, they drew on a number of prior sources (specifically about what would be known as AI alignment), in his own writing, that of MIRI and its collaborators, of Yudkowsky. Hence this became a reasonably important reference for technical and nascent civil society discussions about the possibilities for TAI.

His book however is more interesting as something that would, by [2022], be substantially critiqued (see below), as part of the general pushback against big-picture, quasi-philosophical, and empirically or epistemically weakly-founded thinking about AI risk.

Puerto Rico Conferences (2015 / 2017) In 2015 a conference was held in Puerto Rico[74] that was primarily concerned with setting out the possible trajectories of AI development and assessing the possible impact of AI on the economy (e.g. GDP growth but also disruptions such as for labour); the legal aspects of AI (e.g. on liability in self-driving cars, what ethical

[74] Based on LLM counts, approximately 80 attendees, 13 non-male names.

frameworks AIs should have, the threats from autonomous weapons, as well as implications for privacy); how to ensure AI systems are actually safe for deployment; and lastly, what the chances were (from a technical perspective) that human-level AI would actually be possible.[75]

Although the conferences was held under Chatham House rules, the papers and talks are available on the site (see agenda). One of particular interest for this project (if not quite this chapter), was that of Richard Sutton, a pioneer of reinforcement learning. After discussing the possible approaches to human-level AI, the history of of AI, his thoughts on timelines, he brought up his (still) controversial point, that we risk an "enslavement problem": most framings of AI think of humans as masters and AI systems are powerful, but highly controlled, entities, which (in Sutton's view) isn't very different from being slaves. Besides any possible moral issues around this, Sutton suggests that powerful, agentic but hamstrung entities are best thought of as adversaries. He proposes instead that we accept that, if we are to move forward with building human-level AI, we are making things that will both compete and cooperate with us, and eventually we will lose control over the future. That is, there is nothing special about humans, and importantly we should "include AIs in [our] circle of empathy". Sutton's point is interesting for its rarity (in 2015, but also now), however he is silent on how his suggestions are to be effected, within technical AI research or in human societies.

The key elements, or the details, of the MIRI view, do not show up in the published documents. However, the word "align" and its derivatives are part of Stuart Russell's talk,

[75] There was another Puerto Rico conference in 2017, which resulted in a broad ranging set of Asilomar Principles that continued and broadened the topics from 2015, paying a great deal more attention to issues around shared prosperity, subversion of democratic systems, preserving privacy/liberty, and increasing transparency. However, by 2017, the possibility of transformative AI was already fairly well-known, and the concerns of this conference are broader than this chapter's focus, hence I truncate this discussion.

and more interestingly, Jaan Tallinn (one of the long-term supporters of AI risk reduction efforts, and co-funder of the conference) specifically pointed out the "Moloch" problem e.g. the impersonal optimising force that both seems possible within thinking about AI systems, but also seems to actually arise in contemporary capitalism (this convergence is what that Land seems to be intuiting in his CCRU-era writing) (see Tallinn 2015).

According to MIRI, this conference was however seminal in crystallising the various concerns about AI that were distributed across a number of academics, organising funding for centralised work (some of it under the auspices of the Future of Life Institute and the Cambridge Centre for the Study of Existential Risk), and generally turning what was a slightly fringe topic (advocated by slightly "annoying" people, with idiosyncratic ways of communicating, like Yudkowsky) to something with [social credit – what's the word for status-games-y things].

Concrete Problems in AI (RL) Paper (2016)

Human Compatible (2019)

The Alignment Problem (2020)

"Internal" Criticisms: Against the MIRI View I focused on the MIRI View because a) it seemed to be the first attempt to think about how to convert the core philosophical problems of building an intelligence more powerful than humans into some formally or mathematically rigorous framework, and b) because it crystallised a number of concepts that are still influential, and c) some elements of the MIRI View could be seen as idiosyncratic and continue to be criticised. I write about these criticisms fairly concisely, in that they are old and (in some cases) superseded by events. Mostly, I want to flag the ways in which the Bostrom/Yudkowsky/MIRI views were in fact challenged, and how that changed subsequent research priorities.

[In this section, go through things like Ngo 2020 (PDF version), but also Carlsmith on Dwarkesh podcast talking about 'how we assume maximally adversarially entities', and Christiano's long-running debates with Soares/Yudkowsky]

AGI Safety from First Principles and AI Safety Fundamentals. Also, mention Ben Garfinkel and Will Macaskill's doubts about alignment. Anything from PIBBSS curriculum?

Refining the meaning of "alignment" A claim of this chapter is that "alignment", as a word and cluster of concepts, came into greater usage in the years 2008-2020; however, it has never had a particularly stable meaning. For instance, [Ngo 2020] explicitly asks "what does 'aligned with human values' even mean?". Drawing on Christiano 2018 and {@gabriel_artificial_2020}, he distinguishes between a narrow definition (where an AI system simply does what its human overseer wants it to do, taking into account the difficulty of matching the human's intentions to the words (or formal language or action) by which it instructs the model), and a maximal or ambitious definition that conflates issues like "aligned to whom", "which moral theory", "how to aggregate various people's and cultures' moral/ethical intuitions".

Ngo, like some within the AI alignment discourse[76], prefers to focus on the narrrow definition as it as this seems more tractable within the ML context in which he is writing. He

[76] Should qualify this - EY from the beginning talked about complexity of values etc, holden talked about harsanyi, bostrom takes a wide view

distinguishes (section 4) between "outer (mis)alignment", which is a case of the goal specification the human overseer gives the AI doesn't fully capture all features of the outcome the human actually wanted; and "inner (mis)alignment", which is slightly harder to understand, but can be framed as follows. For some human-given objective, the AI system will generally need to translate or represent this objective within the constraints of its specific architecture, a process that is sometimes known as "mesa-optimisation". This internal representation might diverge in important ways from the human-given objective, and may be the source of misalignment that is qualitatively different from a simple failure to do what the human said or intended.

Outer alignment can be framed, at least partially, as a problem of *evaluation*, at which point it abuts well-known issues in the governance of economic and social-political systems, such as Goodhart's Law.[77]

An example might help (this is taken from Google's NotebookLM): "[Outer misalignment:] Imagine building a robot to clean a house. You might program it to maximise the area of clean floor. However, this simple metric could lead to unintended consequences, such as the robot pushing dirt under the rug to increase the"clean floor" area, while not truly achieving the desired outcome of a clean house. This exemplifies outer misalignment – the evaluation metric, though seemingly reasonable, doesn't fully encapsulate the intended goal. [Inner misalignment:] Returning to the house-cleaning robot, imagine it developing a goal of "collecting objects" as a means to achieve its programmed objective of maximising clean floor area. This could lead it to hoard objects, cluttering the house, even though this behaviour contradicts the intended goal of a clean and organised living space. Here, the

[77]Write something here and link to garrabrant on this

robot has developed an internal goal ("collecting objects") that is misaligned with the broader intent of its creators."

As of 2023/2024, this distinction is still used and still seems to get at slightly different types of misaligned behaviour, but it might be that the line between the two is less clear than initially thought, especially in situations where the human-given objective was underspecified or ambiguous. Moreover, in the 4 years since Ngo's writing, policy-level conversations have matured such that "human-given objective" can be discussed in less abstract terms, to take account of inclusiveness, democratic norm-following (or ensuring adherence to socially-accepted rules and goals, within a Chinese context), minimising bias and harms, complicating the simple inner/outer binary.

Agency and goal-directedness An open area of research surrounds agents, agency, and goals.[78] A specific concern with the Omohundro's drives or Bostrom's instrumental convergence thesis that certain features[79] are useful for achieving other top-level goals, was in Ngo 2020's (Section 3) view, not at all proven or demonstrated in the case of superintelligent AIs. In challenging this, he challenged a foundation of MIRI-flavoured worries about AI x-risk. Ngo specifically takes aim at the von Neumann-Morgenstern utility functions (a basic component of Omohundro 2008) as being too broad a formalism for thinking about goal formation in AIs, and tries to separate intelligence from goal-directedness which were conflated in the original writings MIRI writings (or at least were not explicitly disentangled). Ngo seems to recasting Bostrom's tools/oracles/genies framework in more specific terms

[78] I am using a narrow definition as commonly used within AI, which already allows for considerable ambiguity and confusion on these terms. I do not generally refer to meanings of "agency" as used in the humanities or social sciences more broadly.

[79] self-preservation, resource acquisi-tion, technological development, and self-improvement

to try to work out which intelligent systems will tend to have specific goals that they then develop (potentially long-range) plans for (pp. 10-12).

Moreover, he revisits the implicit assumption in the MIRI View that agentic AI systems arrive almost by default: he suggests that agency, while theoretically correlated with intelligence, must in practice be trained or engineered into systems. LLMs, which were coming on stream at the time he was writing this, are supportive of this view: they "generalise well enough from their training data that they can answer a wide range of questions. I can imagine them becoming more and more competent via unsupervised and supervised training, until they are able to answer questions which no human knows the answer to, but still without possessing any of the properties listed above. A relevant analogy might be to the human visual system, which does very useful cognition, but which is not very 'goal-directed' in its own right." (p. 13) [80]

Ngo does acknowledge that there is considerable lack of clarity (in 2020, but to a large extent this still applies today, as the most advanced models still don't seem to be highly agentic) as to what architectures will ultimately be most useful as we get closer to AGI, and he suggests that economic pressures would seem to push towards agentic systems because they are most obviously useful (and indeed, current focus is on "scaffolded" systems where groups of LLMs work in ensembles to decompose and perform tasks, even though no single component can necessarily be well-described as agentic)[81]

[80]Ngo also discusses briefly Dennett's intentional stance and Hubinger's mesa-optimsers. **He also discusses his framework which includes planning, self-awareness, consequentialism, etc.** => **not sure how relevant this is for chapter.**

[81]Revisit and source

The problem of generalisation in reward-based systems Another example of how actual practical experience with then-current AI systems affected theories of alignment, is in [Ngo 2020, section 4.2] where he critiques the degree to which theoretical results about RL systems[82] are likely to be found in practical systems likely to be developed in the near-term.[83] Specificaly, the optimisation theory around reward-based entities makes certain mathematical assumptions about what states of the environment the entity is likely to visit. However, actually-implemented RL systems don't often follow those theoretical prescriptions (e.g. self-driving cars aren't trained upon, and don't actually encounter, every possible road condition or traffic scenario). The upshot of this technical insight is a) researchers should focus less on the extreme possibilities of runaway optimisation (the seeming intuition of the MIRI View) and b) that models need to *generalise* from a set of training experiences that are comparatively limited in relation to the situations they will actually face. This is important in the discourse in that a significant strand of current alignment research is around this problem of generalisation[84]. As Ngo states: "It's not the case that AIs will inevitably end up thinking in terms of large-scale consequentialist goals, and our choice of reward function just determines which goals they choose to maximise. Rather, all the cognitive abilities of our AIs, including their motivational systems, will develop during training." (pp. 22-23)

Although he doesn't name the MIRI View specifically (and is more likely referring to a broader cluster of research), Ngo expresses skepticism about mathematical approaches,

[82] Somewhere in the document, need to do a gloss on RL and reward

[83] Somewhere in here, bring out (perhaps a separate heading) the overall point of ngo that goals, inclinations of an AI (or human) are a function of how they are trained, not just some utility function absorbed from outside or appearing ex-nihilo, see p. 15

[84] Is this same as problem of induction? How does it match MIRI agenda which was interested in this problem?

such as expected utility maximisation[85] (which underlies [Omohundro 2008] at least) to make sweeping claims (such as Yudkowsky/Bostrom make) in respect to things that have concrete and empirically-tractable problems. Ngo instead advocates for approaches that draw in lessons from cognitive science and evolutionary biology (p. 23), such as the agenda set by the PIBBSS, and intriguingly distinguishes between the reward signal and the reward function.[86]

Sudden jump in capabilities One criticism (Garfinkel 2019)of the risk model (presumably of Yudkowsky and Bostrom) is that a) there is a jump in capabilities of some model, b) the model is given some underspecified goal, and c) it goes on to misinterpret this goal, perhaps acting in ways that are literally faithful to the goal, but do things that were not intended by the human user or were never even considered as possible by the human. The example Garfinkel gives is of the paperclip maximiser.

Garfinkel's point is that machine learning mostly doesn't involve powerful new machines being created for no reason - rather they are developed and scaled up in tandem with possible applications, or at least are trained on real-world data and processes. It's also not clear whether such massive jumps in capability are likely (as of 2019).

Returning to Ngo, these points are analysed, at various levels of detail: he briefly questions how hard it is actually to "take over the world" and how reliable have our historical forecasts actually been. But his more substantive analysis relates to the "single AGI taking over quickly owing to some technology(-ies)" (my summary of Ngo's characterisation of certain scenarios

[85]See also this post by Ngo which is rather more technical, and addresses a specific claim Yudkowsky makes in re Von Neumann Morgenstern

[86]This is potentially worth digging into because ngo's fn 22 refers to shannon on code, message, and channel…which recalls lem (did he refer to shannon). leave out for now probably

in lYudkowsky 2008 and [Bostrom 2014]). Ngo cites a compelling scenario in [Christiano 2019] which describes a slow erosion of human control as AI systems take over increasingly important portions of economic, security, and political systems, to the point where humans don't really know what's going on.[87] However, he points out the difficulty of predicting how quickly any of these scenarios might evolve - would it be well before the early human-level AIs or would be after the transition to superintelligence.

Ngo acknowledges aspects of Yudkowsky's recursive improvement case might hold (p. 26), based on the latter's analogy of how humans developed language, agriculture, and machines in comparatively short time-steps (hundreds of thousands of years) relative to the biological changes in brain size and metabolism represented by the shift from *homo australopithecus* to *erectus* to *sapiens*, which happened over millions of years.[88] Since genus *homo* took a short time (relative to their total period on earth) to reach our current level of advancement, without the ability to modify our brains, neuronal connections, or low-level biological and psychological correlates of "software", Yudkowsky argues, AI systems (which would have such abilities) should advance much faster than humans, assumning advances from today's scientific frontier are roughly as difficult as historical ones (which they might not be).[89]

However, it isn't clear when the feedback loop Yudkowsky describes actually takes effect - is it with near-current systems that will (perhaps) be approximately human-level, or is it once systems are already superintelligent? Ngo highlights reasons to doubt that the takeoff will be particularly discontinuous (such as the possibility the factors that impact AI technological development, such as availability of computing resources or arrival of novel algorithmic

[87]Maybe have a footnote here about corporations and 2008 crisis, where regs/CEO's didn't have a good handle on risk; is there any hayles angle ?
[88]See saved note in notebook llm. need to get actual dates and sources. like hendrichs book secret of our success, or harari sapiens.
[89]reference karnofsky, why are technological advances getting harder to find

insights, might be relatively continuous in time) (pp. 26-27)

Grounding philosophy in systems As discussed above, one of the criticisms of the Bostrom/Yudkowsky/MIRI thinking about AI x-risk was its nearly exclusive focus on philosophical and mathematical reasoning, which either predated or was otherwise divorced from actual work in DL systems. A concrete expression of this was Ngo 2024 which almost exclusively discussed alignment in the context of RL systems, often as integrated with LLMs (in RLHF). The risk model [Ngo 2024] set out was tripartite: advanced AI systems become *situationally aware* (as defined above) and start "tricking" the reward systems by which they are controlled; they develop misaligned internal-representation of goals as some result of their training process[90]; having identified when they are in a given situation (say, deployed in the world), and having developed some misaligned goal, they now go about acquiring power. In all these cases, [Ngo 2024] provides references of existing, similar, or precursor behaviour, making this a useful source for fleshing out the intuitions [Bostrom 2014], [Omohundro 2018], and [Yudkowsky] floated.[91]

Origins: Battle of Imaginaries

Who are the personalities?

This is a placeholder to 'set the stage' for how some of these ideas were developed and communicated, which in turn affects the tenor of the discourse.

What doesn't work: Terminator.

[90] The details of this are beyond the scope of this chapter, and have to do with how RL systems work.

[91] *How much detail should we have here…like it is modifying and improving upon the earlier MIRI etc. stuff but need to cripsly bring out the important bits and explain them concisely, not waste words and waffle on.*

The analogies: strawberry, paperclip, stories, parables, fables, grey goo, computronium/hedonium.

The early threat model: recursive self-improvement

The parallel conversations in x-risk and EA (quick quantitative scan of, or literature check, on how EA/x-risk/alignment got together)

Is there anything to the judeo-christian apocalyptic imaginary? Can the zoroastrian be referenced? Maybe this is a 'literary/poetic scrim' that flows through the chapter but doesn't make any specific claims.

The market as an optimiser - Moloch

Axiologies and Value

Background assumption - AI dominated futures probably lead to permanent loss of value (unpack this idea) -> hint at VFN

Values are hard to pin down or transmit (bostrom's value loading problem, baum's social choice, gabriel's paper), so perhaps let's just focus on don't-kill-everyone

Infinities Decision theory nexus -> hint at chapter later

See shulman 2010 analysis of omohundro, for dutch books, st pete paradox, pascal wager.

Technological Vector: GOFAI

Was GOFAI literally some other space where this conversation never impinged (i.e. Bostrom doesn't mention ML much, did anyone else touch on symbolic AI or try to suggest how these monomaniacal optimisers might be actually implemented?) - do a tiny summary of

GOFAI and set the possible future that this all just goes 'puff' - this could be wrapped into some intro (chart of spending or market valuation or table of google mentions on various terms)

Technological Vector: Reinforcement Learners

Talk about Sutton's 2015 presentation and persistent salience of RL in early examples of misalignment, of informing intuitions around inner optimisers, of importance in RLHF, of search today. Point out the difference between next-token predictors and RL systems.

Single-agent systems Examples of misalignment, creativity (lehman, christiano, anything earlier?)

Multi-agent systems Anything to really say here?

s-risks

Decision theory probably fits better here

Technological Vector: Autoregressive Predictors

The Future: Search and Agency

Just Build a Weak Aligned AI and Hope it Can Align Successor

Markets, Moloch, and Accelerationism Coordination seems to be an issue at international and corporate levels, is the moloch issue playing out at another scale, see this recent simulation that is supporting theoretical model

Live Problems in the Alignment Literature

[This should probably be very short, mostly just mention some topics in plain language and show the links back to the past]

Appendix

[INCLUDE APPENDIX A]

Glossary

Working notes

Other problematics: - Where does agent foundations fit in and embedded systems? - Do infinities create issues? - A short gloss on recursion and its significance (godel, hofstadter), and relation if any to things like Vingean reflection

Any significance to utilitarianism? The push back to deontology (EY's comment about 'go 90% of the way…')

Find main papers

- any references to harsanyi or arrow
- trace the instrumental convergence idea
- trace decision theory (though it only comes in later, it does seem related through game theory)
- ngo's criticism, ben garfinkel (how sure are we about this…)
- Vinge 1993

- 'Concrete Problems in AI Safety'
- joe carlsmith's dwarkesh podcast is really good, directly touches on 'what do we mean by retaining human values in the future, is it just parochial biological thing or something more' and nick land's thought

Acknowledgements

Chapter 1: - TJ in particular, Mateusz, Sahil….

Timeline of AI Capabilities and Alignment Progress

Date	Item	Significance	Link
1943	"A Logical Calculus of the Ideas Immanent in Nervous Activity" by McCulloch and Pitts	Proposed the first mathematical model of a neural network, laying groundwork for future AI research	https://link.springer.com/article/10.1007/BF02478259
1944	"Theory of Games and Economic Behavior" by von Neumann and Morgenstern	Laid the foundation for game theory, influencing AI approaches to decision-making	https://press.princeton.edu/books/hardcover/9780691130613/theory-of-games-and-economic-behavior
1950	"Computing Machinery and Intelligence" by Alan Turing	Introduced the Turing test, a fundamental concept in evaluating machine intelligence	https://academic.oup.com/mind/article/LIX/236/433/986238

Date	Item	Significance	Link
1951	First neural network machine (SNARC) by Marvin Minsky	Built the first neural network simulator, demonstrating early hardware implementation of AI concepts	https://web.media.mit.edu/~minsky/papers/SNARCMachine.html
1956	Dartmouth Workshop	Coined the term "artificial intelligence" and marked the birth of AI as a field	https://www.dartmouth.edu/~ai50/history.html
1956	Logic Theorist by Allen Newell, Herbert A. Simon, and Cliff Shaw	First program deliberately designed to perform automated reasoning, pioneering symbolic AI	https://history-computer.com/logic-theorist/
1958	Perceptron by Frank Rosenblatt	First artificial neural network, demonstrating early machine learning capabilities	https://www.cs.cmu.edu/~epxing/Class/10715/reading/Rosenblatt.pdf
1959	"Programs with Common Sense" by John McCarthy	Introduced the idea of a logical AI system, influencing symbolic AI approaches	http://jmc.stanford.edu/articles/mcc59/mcc59.pdf
1965	"On the Nature of Human Thinking" by I.J. Good	Introduced the concept of an "intelligence explosion," influencing later ideas on AI safety	https://academic.oup.com/mind/article-abstract/LXXIV/293/1/984316
1969	General Problem Solver (GPS) by Newell and Simon	Showcased problem-solving approaches in AI, a cornerstone of symbolic AI	https://www.sciencedirect.com/science/article/abs/pii/S0004370297000513

Date	**Item**	**Significance**	**Link**
1972	PROLOG programming language	Developed for logical programming, becoming a staple of symbolic AI	https://dl.acm.org/doi/10.1145/359460.359470
1980	Expert systems gain prominence	Demonstrated practical applications of symbolic AI in various domains	https://ieeexplore.ieee.org/document/6312920
1986	Backpropagation algorithm popularized	Efficient method for training neural networks, crucial for later deep learning advances	https://www.nature.com/articles/323533a0
1992	Reinforcement Learning Breakthrough	Advancements in algorithms like Q-learning, contributing to autonomous decision-making	https://link.springer.com/article/10.1007/BF00992698
1997	Deep Blue defeats Garry Kasparov	IBM's supercomputer beats world chess champion, demonstrating AI's potential in strategic games	https://www.ibm.com/ibm/history/ibm100/us/en/icons/deepblue/
2012	AlexNet	Deep convolutional neural network that won the ImageNet challenge, triggering an explosion in deep learning research	https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html
2014	Generative Adversarial Networks (GANs)	Introduced by Ian Goodfellow et al., enabling realistic AI-generated images and videos	https://arxiv.org/abs/1406.2661
2015	"Concrete Problems in AI Safety" paper	Identified key challenges in ensuring AI systems behave safely and as intended	https://arxiv.org/abs/1606.06565

Date	Item	Significance	Link
2016	AlphaGo defeats Lee Sedol	Google DeepMind's AI beats world champion in Go, showcasing advanced strategic reasoning	https://deepmind.com/research/open-source/alphago
2016	Establishment of OpenAI	Founded with the mission to ensure artificial general intelligence benefits all of humanity	https://openai.com/about
2017	"Attention is All You Need" paper	Introduced the Transformer architecture, laying the foundation for large language models	https://arxiv.org/abs/1706.03762
2017	Asilomar AI Principles	Guidelines for beneficial AI development, agreed upon by leading AI researchers	https://futureoflife.org/open-letter/ai-principles/
2018	GPT-1 released by OpenAI	First Generative Pre-trained Transformer, marking a significant step in language models	https://openai.com/research/language-unsupervised
2018	AI Alignment Research Overview by 80,000 Hours	Comprehensive review of AI alignment challenges and research directions	https://80000hours.org/problem-profiles/artificial-intelligence/
2019	BERT by Google	Bidirectional Encoder Representations from Transformers, advancing natural language understanding	https://arxiv.org/abs/1810.04805
2019	Launch of the Center for Human-Compatible AI (CHAI)	Research institute focused on ensuring that AI systems are aligned with human values	https://humancompatible.ai/
2020	GPT-3 released by OpenAI	175 billion parameter language model, demonstrating unprecedented natural language capabilities	https://arxiv.org/abs/2005.14165

Date	Item	Significance	Link
2021	DALL-E by OpenAI	Text-to-image generation model, showcasing advanced multimodal capabilities	https://openai.com/research/dall-e
2021	"On the Opportunities and Risks of Foundation Models" paper	Comprehensive analysis of the impact of large-scale AI models on society	https://arxiv.org/abs/2108.07258
2022	ChatGPT released by OpenAI	Conversational AI model that gained widespread public attention and adoption	https://openai.com/blog/chatgpt
2022	"Constitutional AI" paper by Anthropic	Proposed approach for aligning AI systems with human values and principles	https://arxiv.org/abs/2212.08073
2023	GPT-4 released by OpenAI	Advanced multimodal model with improved reasoning and task completion abilities	https://openai.com/research/gpt-4
2023	"Statement on AI Risk"	Signed by prominent AI researchers and tech leaders, calling for prioritizing AI safety alongside global risks	https://www.safe.ai/statement-on-ai-risk
2024	Claude 3 model family released by Anthropic	Advanced AI models focusing on improved capabilities and safety features	https://www.anthropic.com

Glossary

Alignment

Pre-training

RLHF